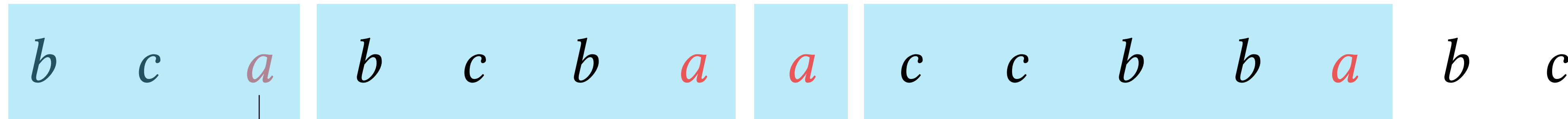
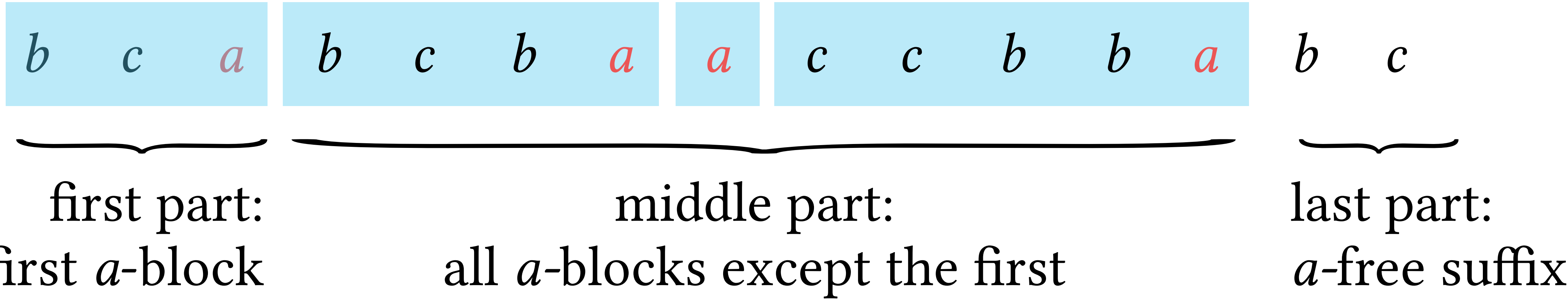
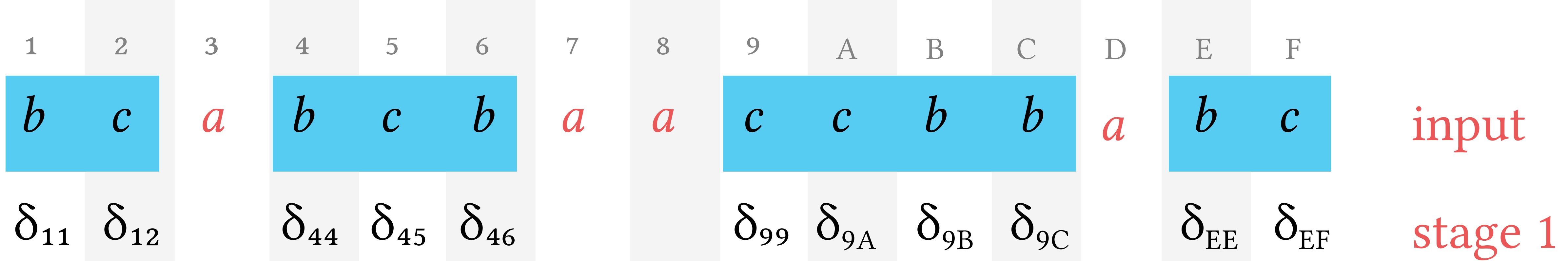


a -block



a -position





1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	
<i>b</i>	<i>c</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>b</i>	<i>a</i>	<i>a</i>	<i>c</i>	<i>c</i>	<i>b</i>	<i>b</i>	<i>a</i>	<i>b</i>	<i>c</i>	input
δ_{11}	δ_{12}		δ_{44}	δ_{45}	δ_{46}			δ_{99}	δ_{9A}	δ_{9B}	δ_{9C}		δ_{EE}	δ_{EF}	stage 1
		δ_{13}				δ_{37}	δ_{88}					δ_{9D}			stage 2

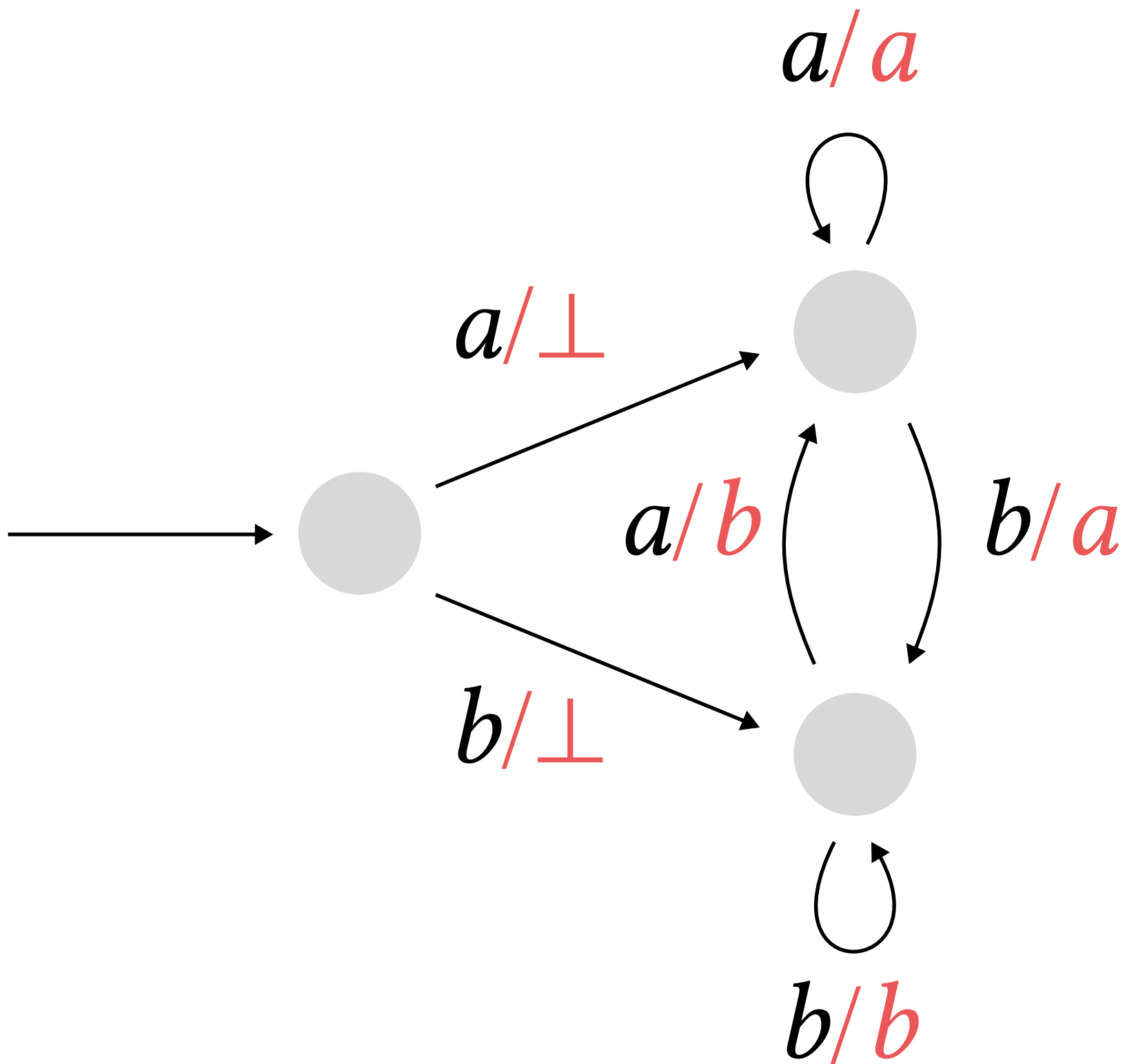
[illegible]

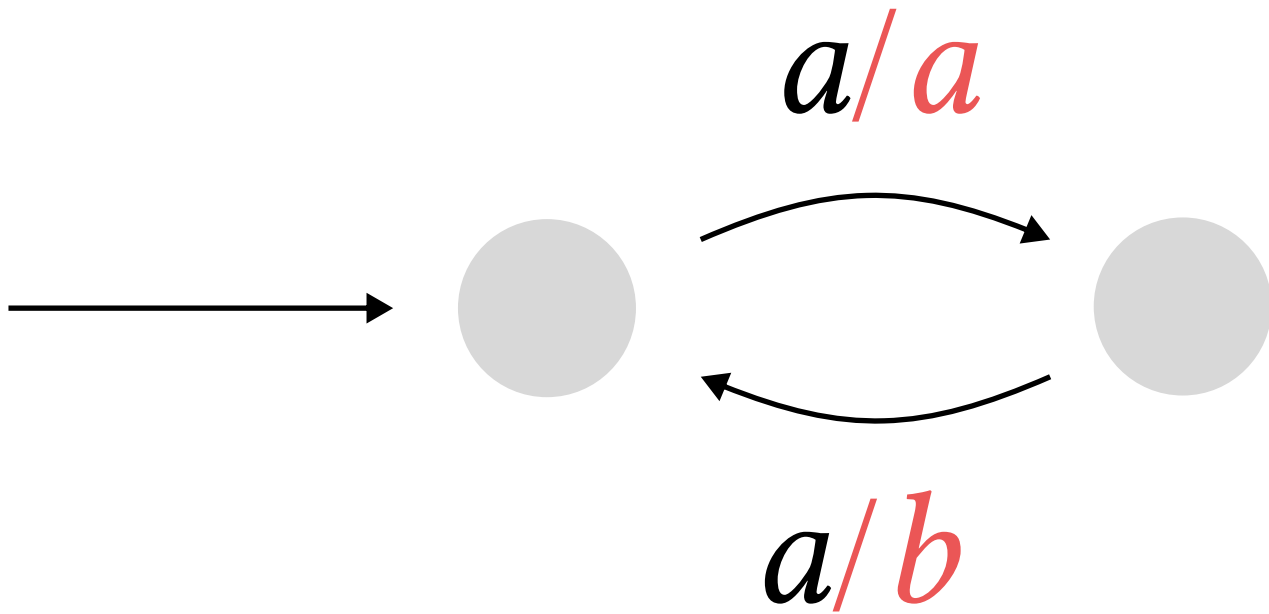
b	c	a	b	c	b	a	a	c	b	input
$\perp$	b	c	a	b	c	b	a	a	c	output

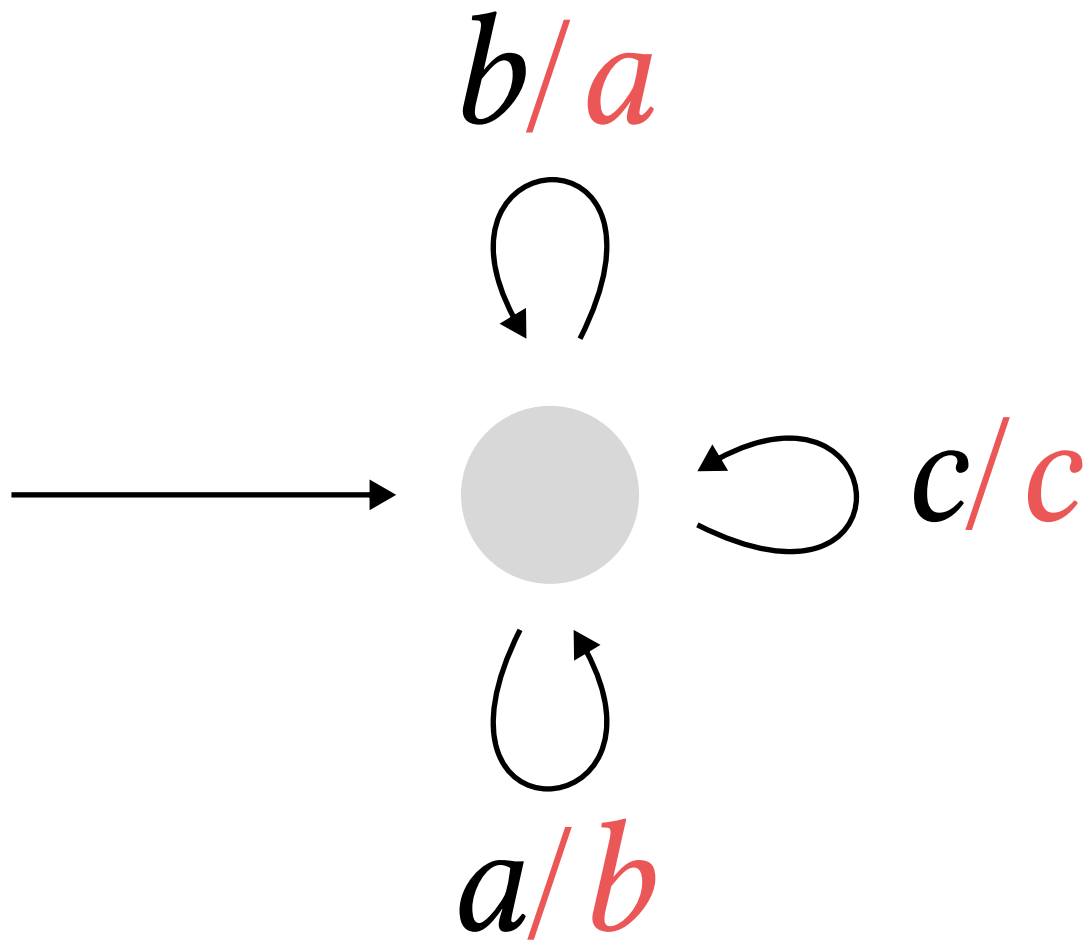
<i>a</i>	<i>a</i>	<i>a</i>	<i>a</i>	<i>a</i>	<i>a</i>	<i>a</i>	<i>a</i>	<i>a</i>	<i>a</i>	input
<i>a</i>	<i>b</i>	<i>a</i>	<i>b</i>	<i>a</i>	<i>b</i>	<i>a</i>	<i>b</i>	<i>a</i>	<i>b</i>	output

1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	
<i>b</i>	<i>c</i>	#	<i>b</i>	<i>c</i>	<i>b</i>	#	#	<i>c</i>	<i>c</i>	<i>b</i>	<i>b</i>	#	<i>b</i>	<i>c</i>	input
Y_{11}	Y_{12}	Y_{13}	Y_{14}	Y_{15}	Y_{16}	Y_{17}	Y_{18}	Y_{19}	Y_{1A}	Y_{1B}	Y_{1C}	Y_{1D}	Y_{1E}	Y_{1F}	stage 1

1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	
<i>b</i>	<i>c</i>	#	<i>b</i>	<i>c</i>	<i>b</i>	#	#	<i>c</i>	<i>c</i>	<i>b</i>	<i>b</i>	#	<i>b</i>	<i>c</i>	input
Y_{11}	Y_{12}	Y_{13}	Y_{14}	Y_{15}	Y_{16}	Y_{17}	Y_{18}	Y_{19}	Y_{1A}	Y_{1B}	Y_{1C}	Y_{1D}	Y_{1E}	Y_{1F}	stage 1
			Y_{13}	Y_{13}	Y_{13}	Y_{13}	Y_{17}	Y_{18}	Y_{18}	Y_{18}	Y_{18}	Y_{18}	Y_{18}	Y_{1E}	stage 2







a_1 a_2 a_3 a_4 a_5 a_6 a_7 a_8 a_9

q_0 q_1 q_2 q_3 q_4 q_5 q_6 q_7 q_8 q_9 run of f

b_1 b_2 b_3 b_4 b_5 b_6 b_7 b_8 b_9

p_0 p_1 p_2 p_3 p_4 p_5 p_6 p_7 p_8 p_9 run of g

c_1 c_2 c_3 c_4 c_5 c_6 c_7 c_8 c_9

input letter

a

q

b

q'

p

p'

c

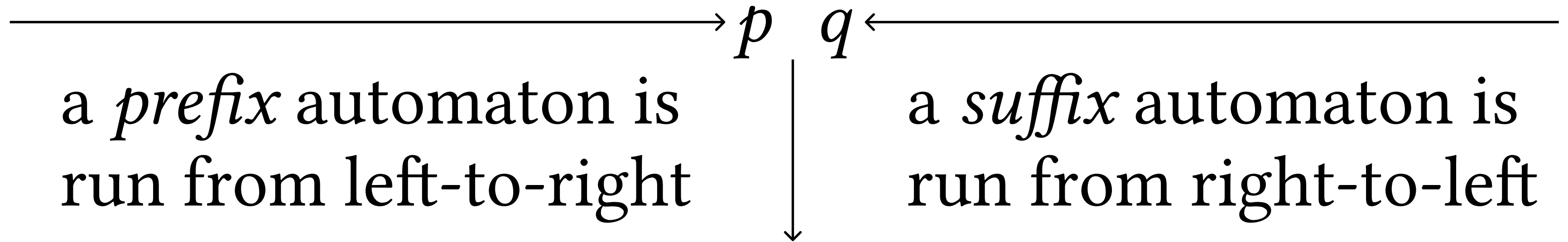
output letter

source state

target state

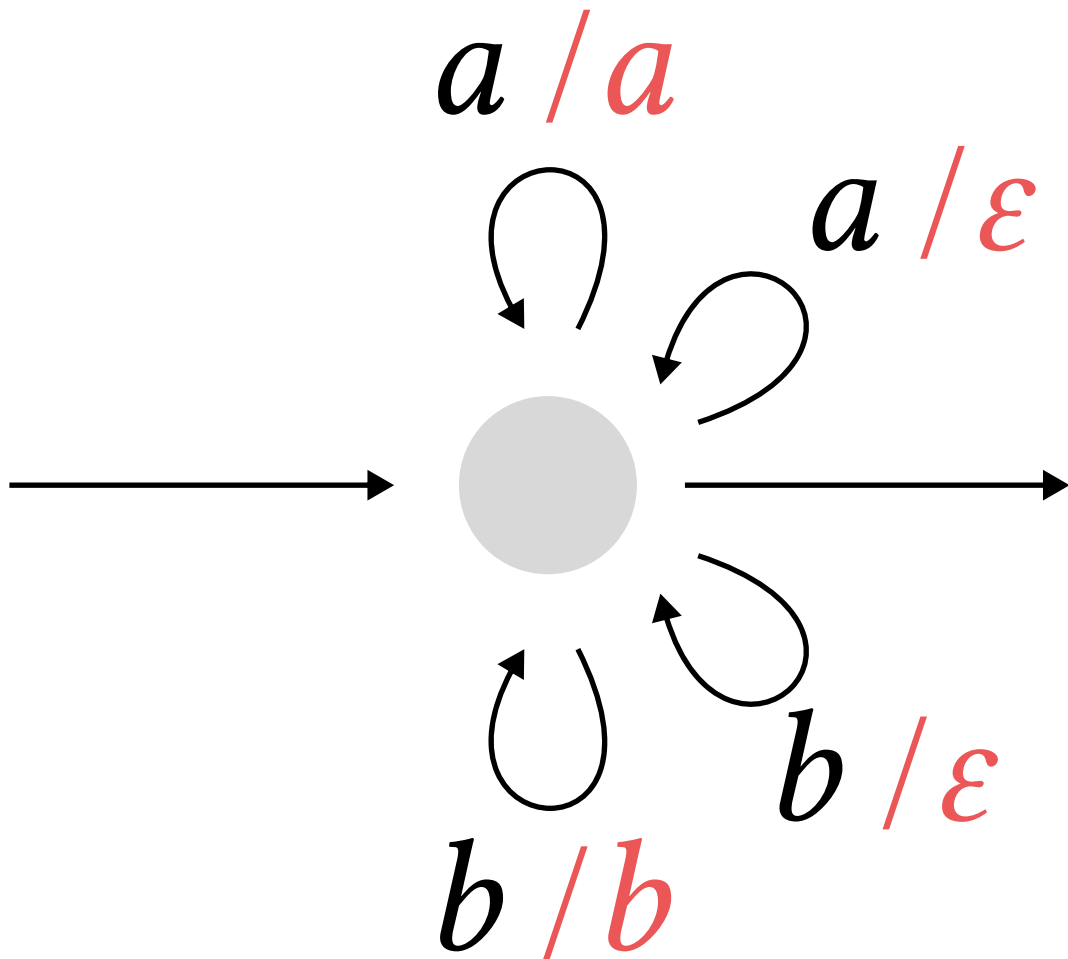
b	c	a	b	c	b	a	a	c	b	input
$\perp$	$\perp$	b	c	a	b	c	b	a	a	output

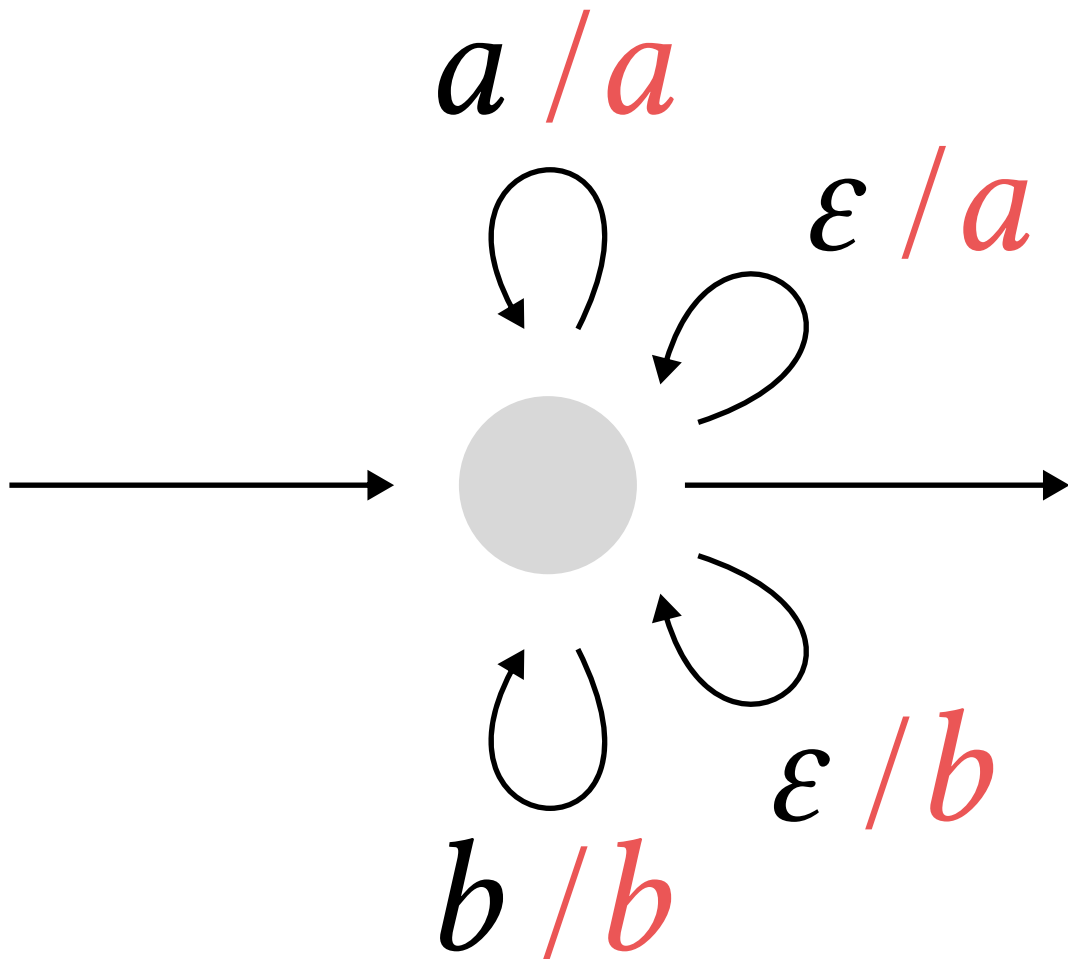
b *c* *a* *b* *c* *b* *a* *a* *c* *b*



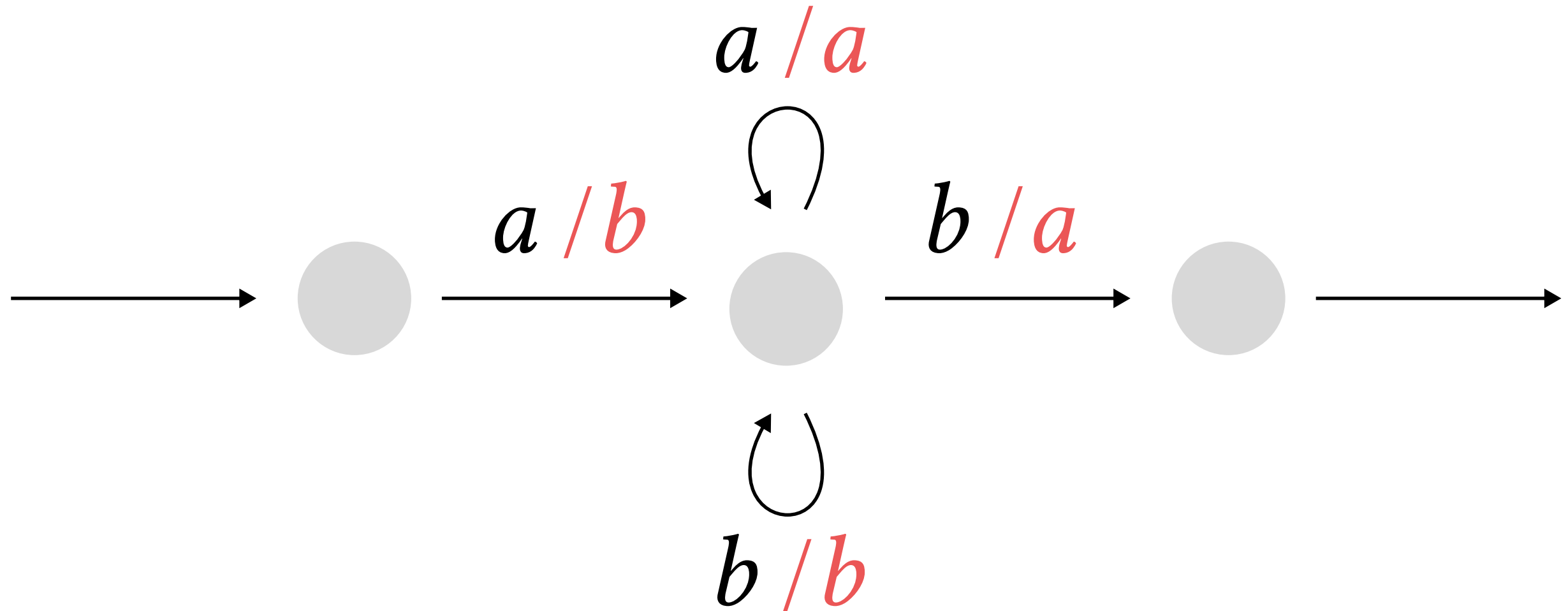
cdd

based on the states of the prefix
and suffix automata, a piece of
the output is produced

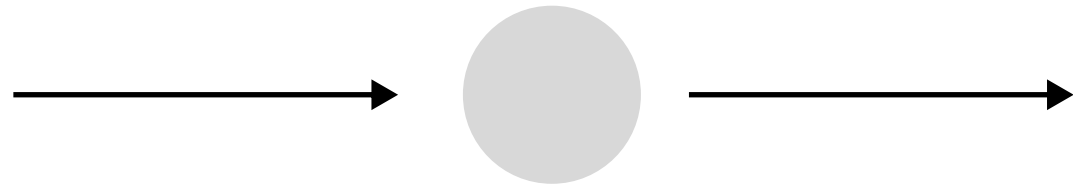




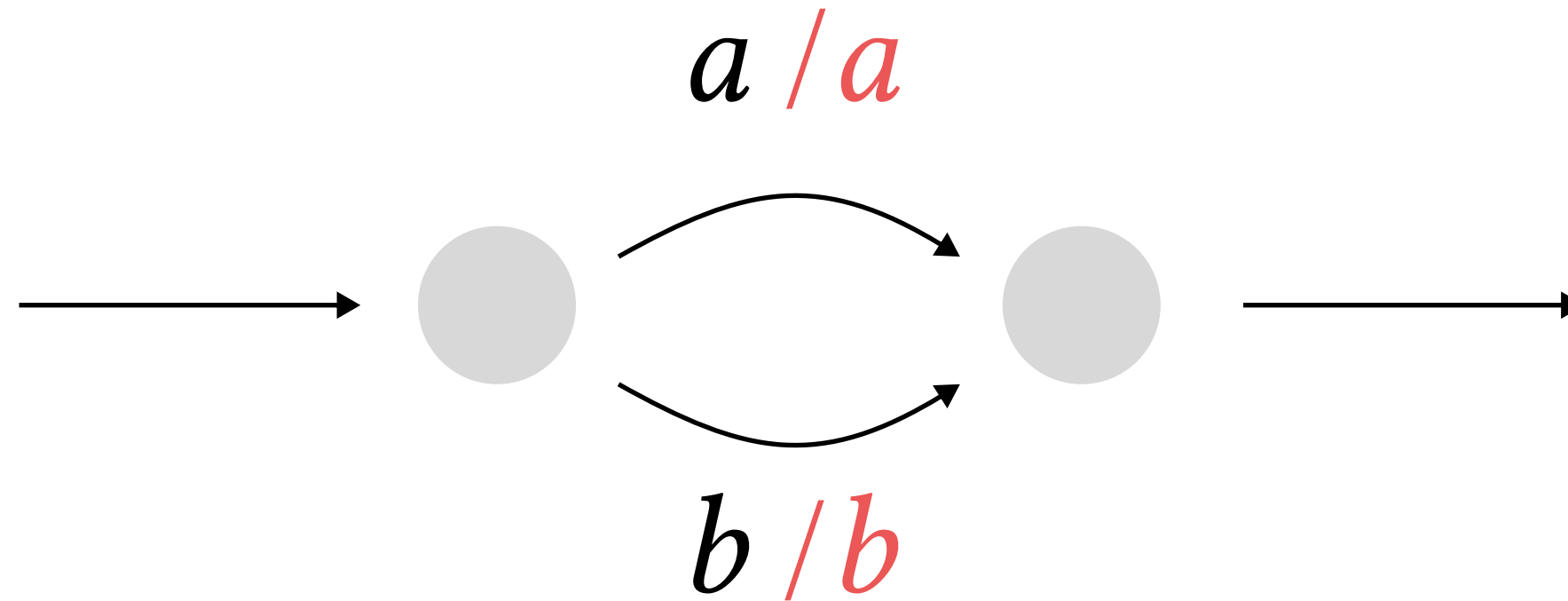
inputs that start with f and end with b



empty inputs

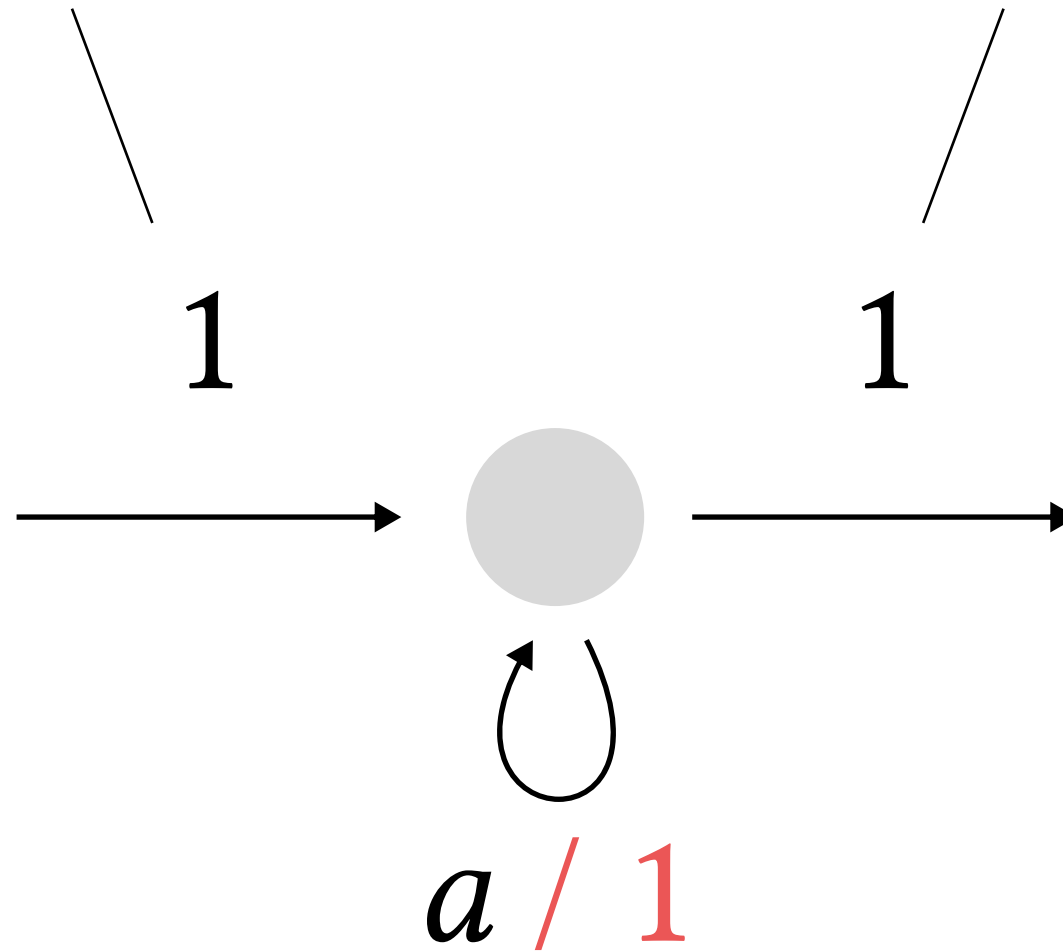


one letter inputs

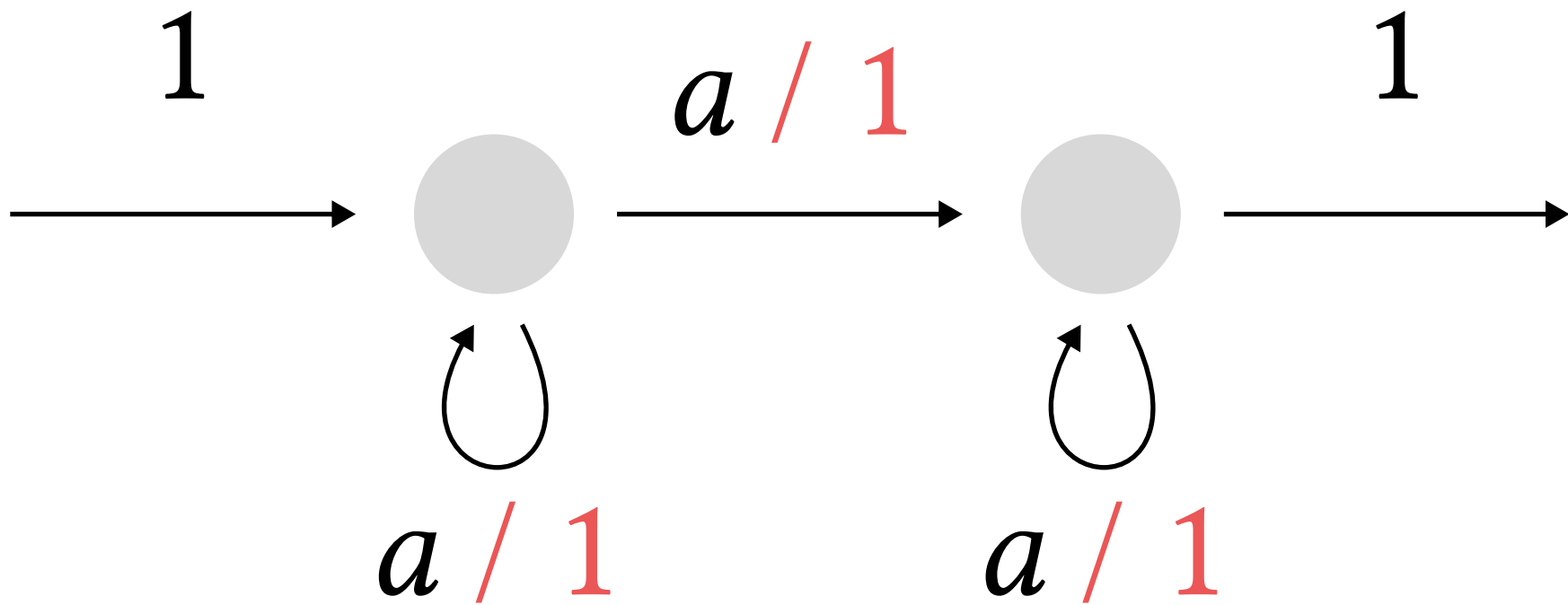


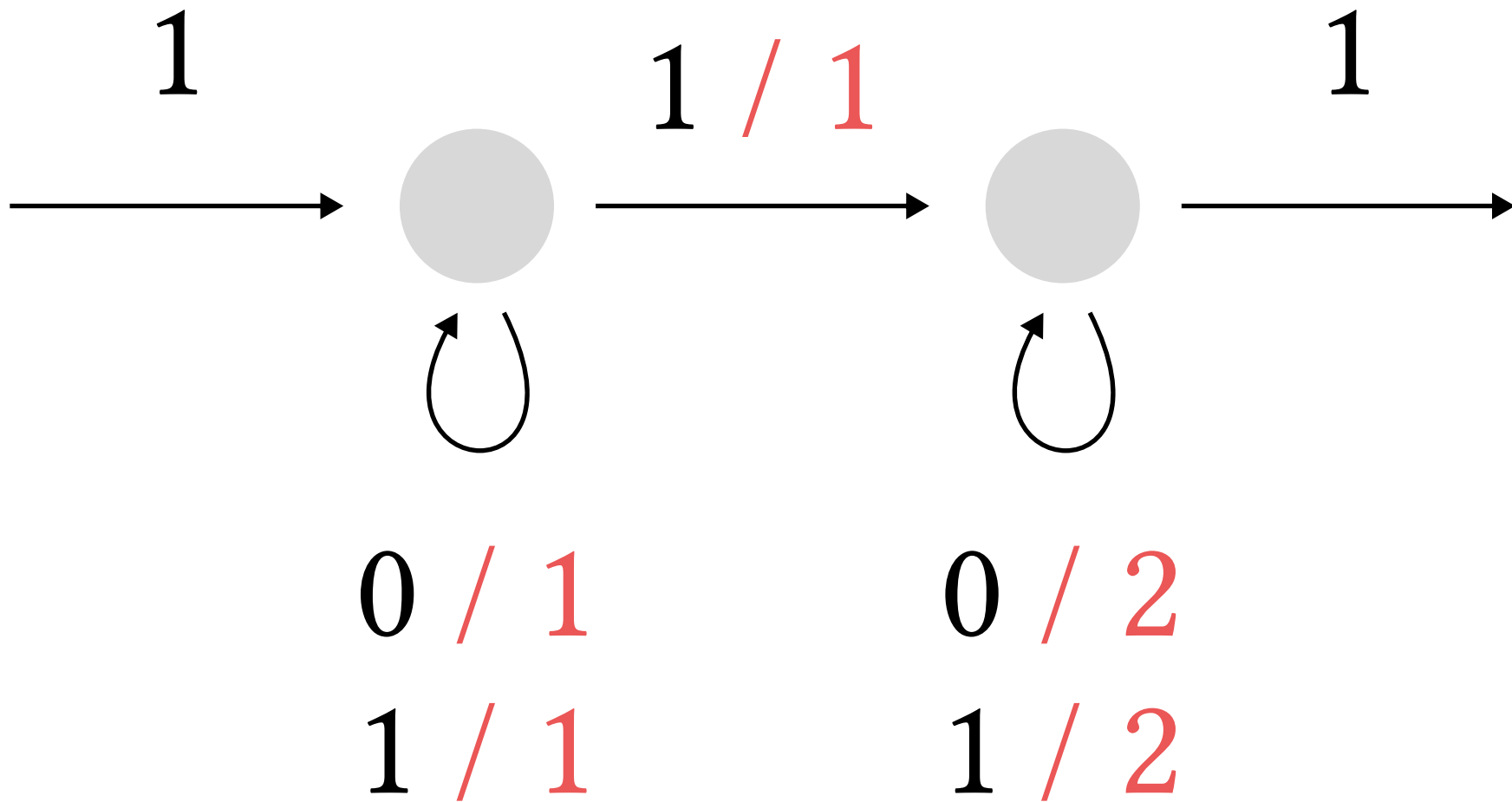
initial weight

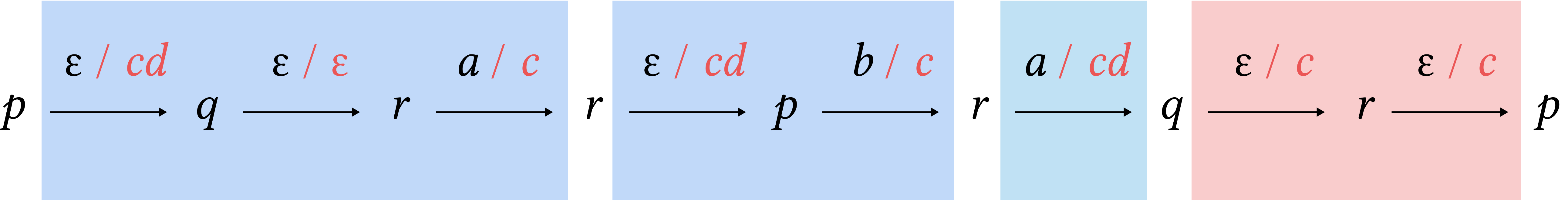
finite weight



transition of weight 1

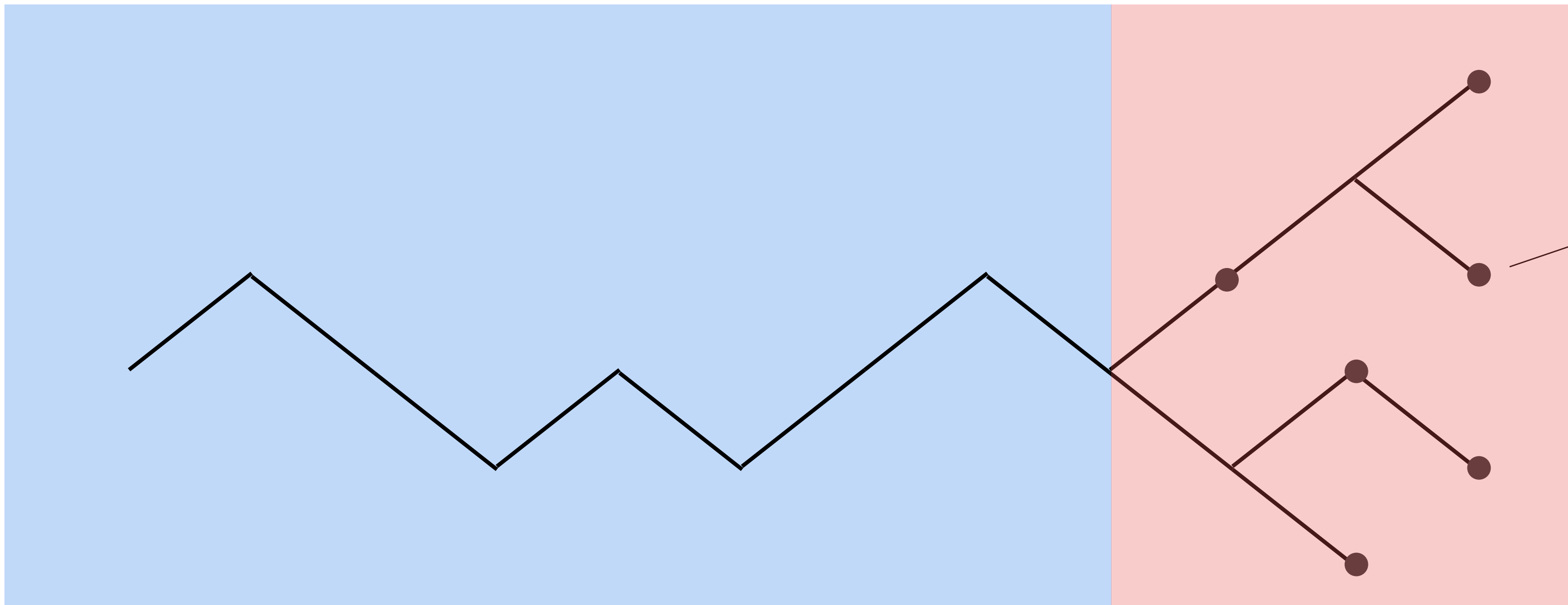




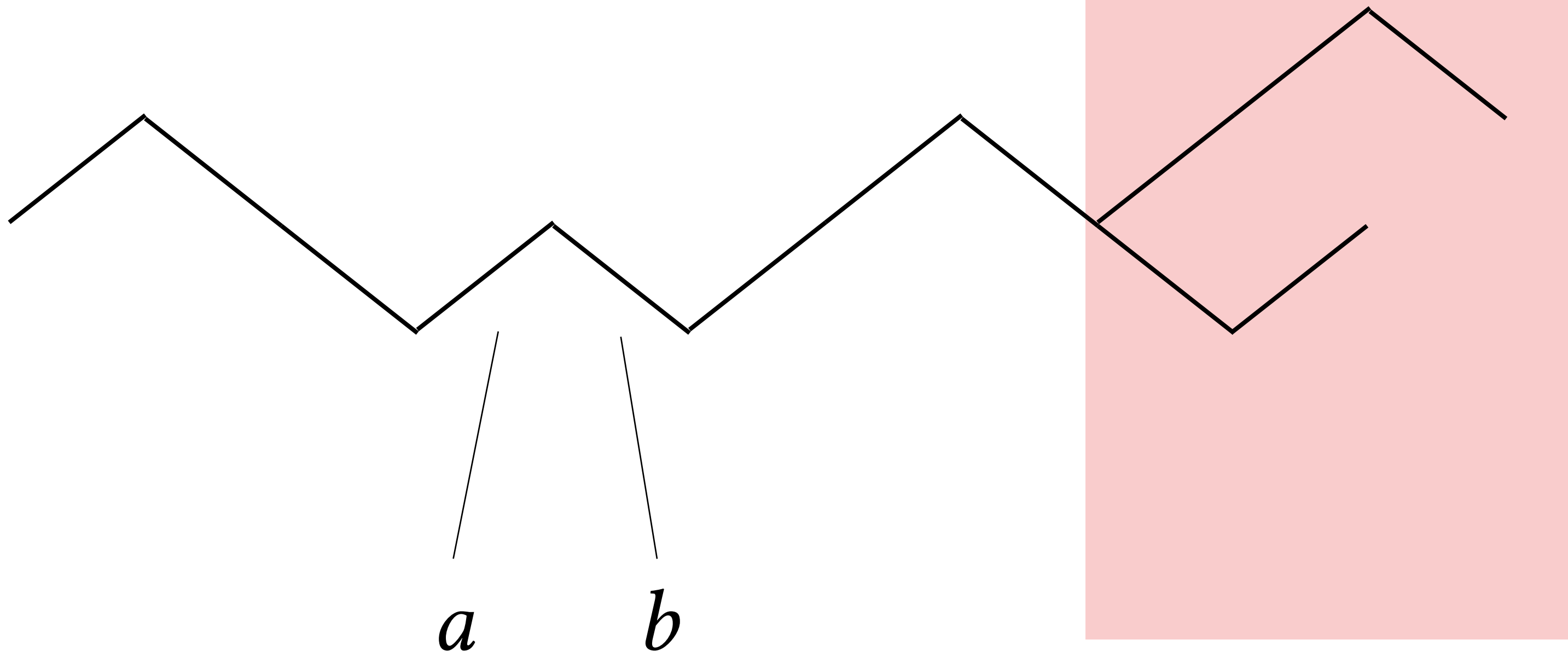


non-branching part

branching part



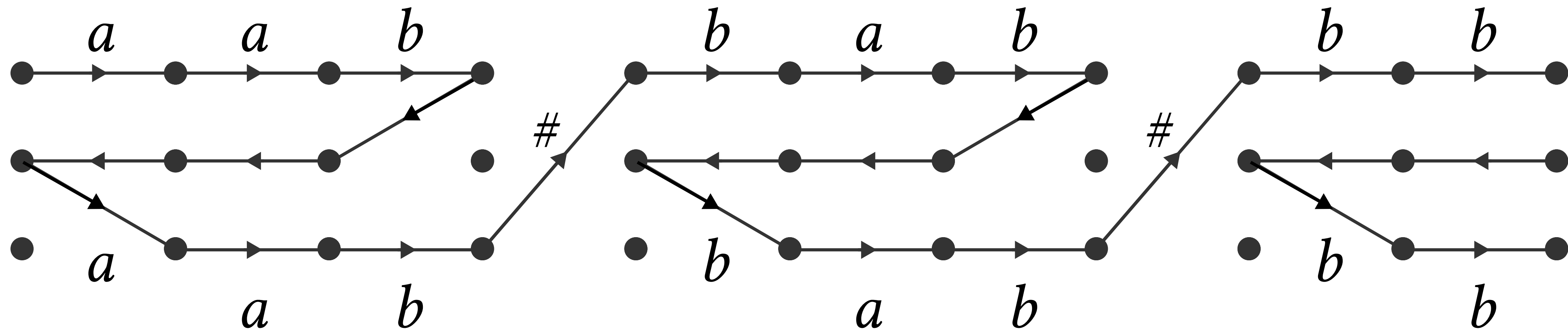
dots indicate outputs
for extensions of w
obtained by adding
at most k letters



directions indicate letters

left distance $k = 3$

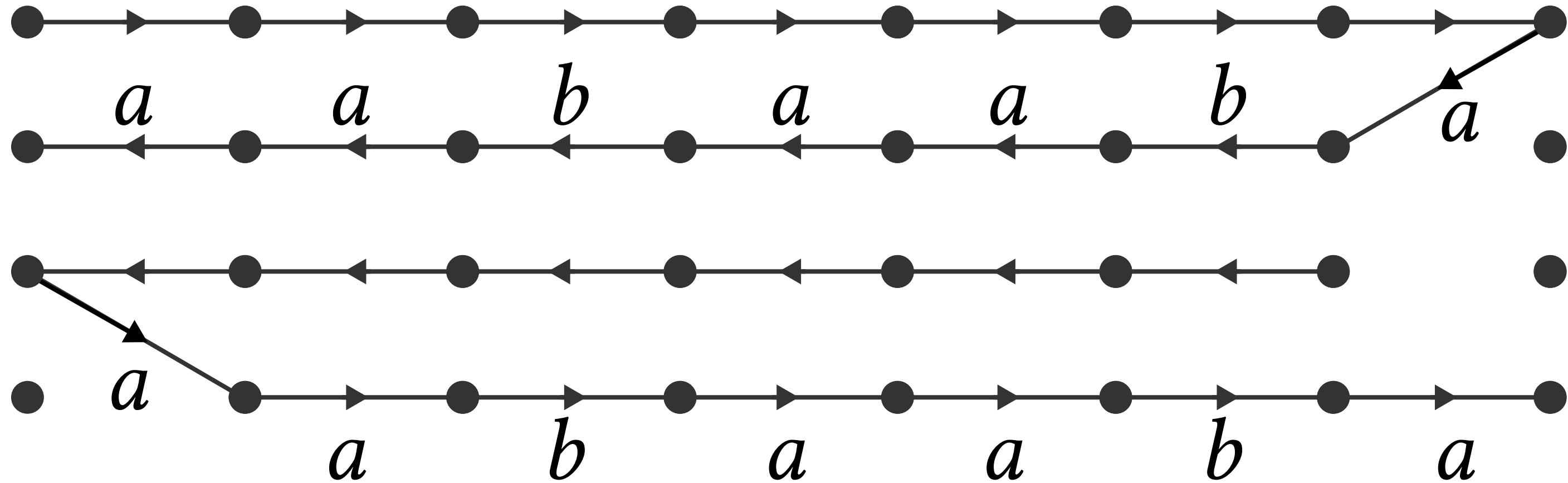
a a b $\#$ b a b $\#$ b b input string



configuration
graph

a a b a a b a

input string



configuration
graph

a b a a a b a a



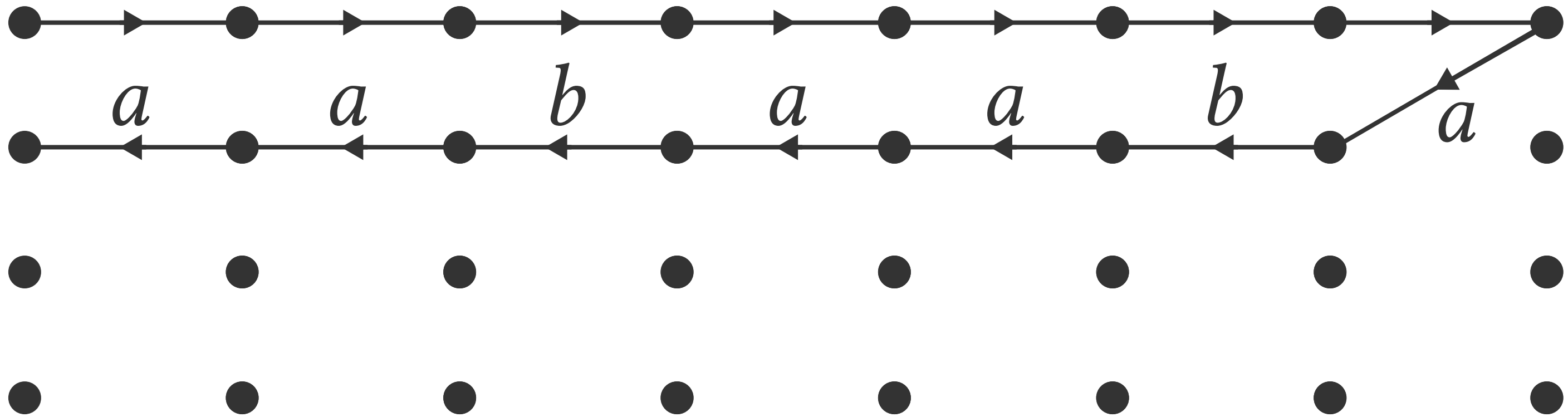
q

a b a a a b a a

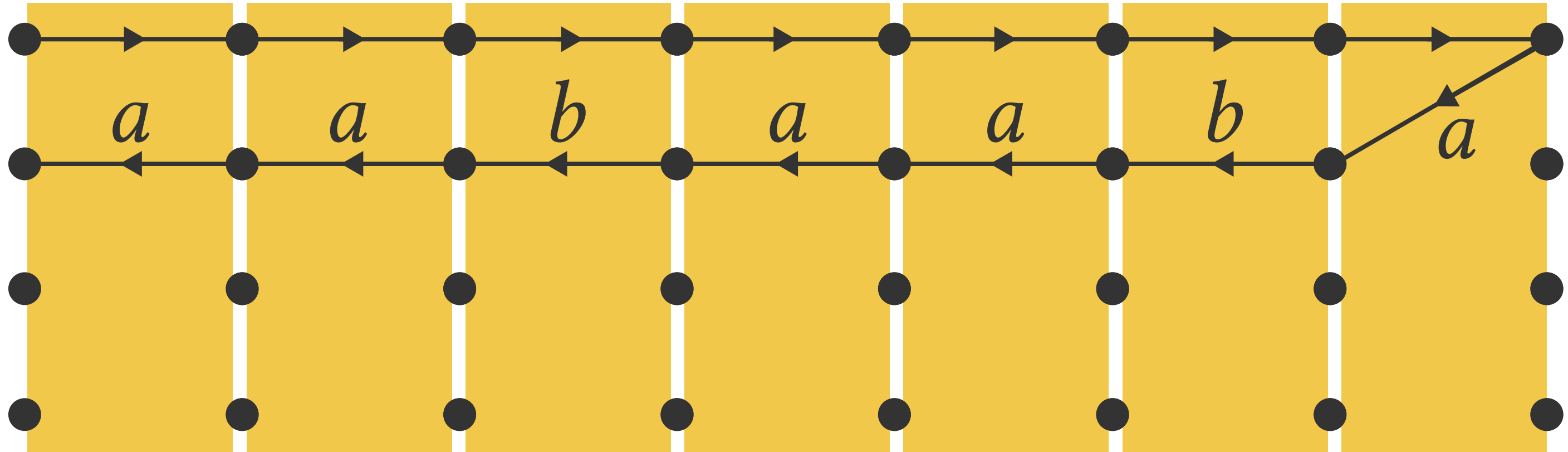


q : state of Mealy machine f

p : state of two-way transducer g

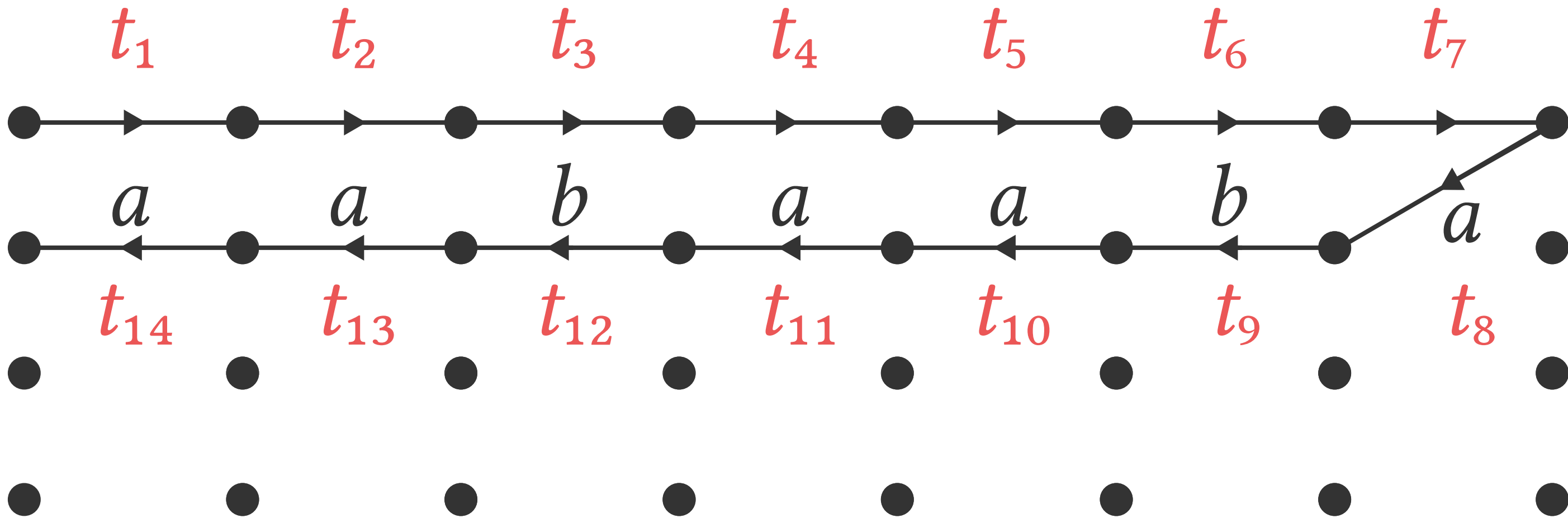


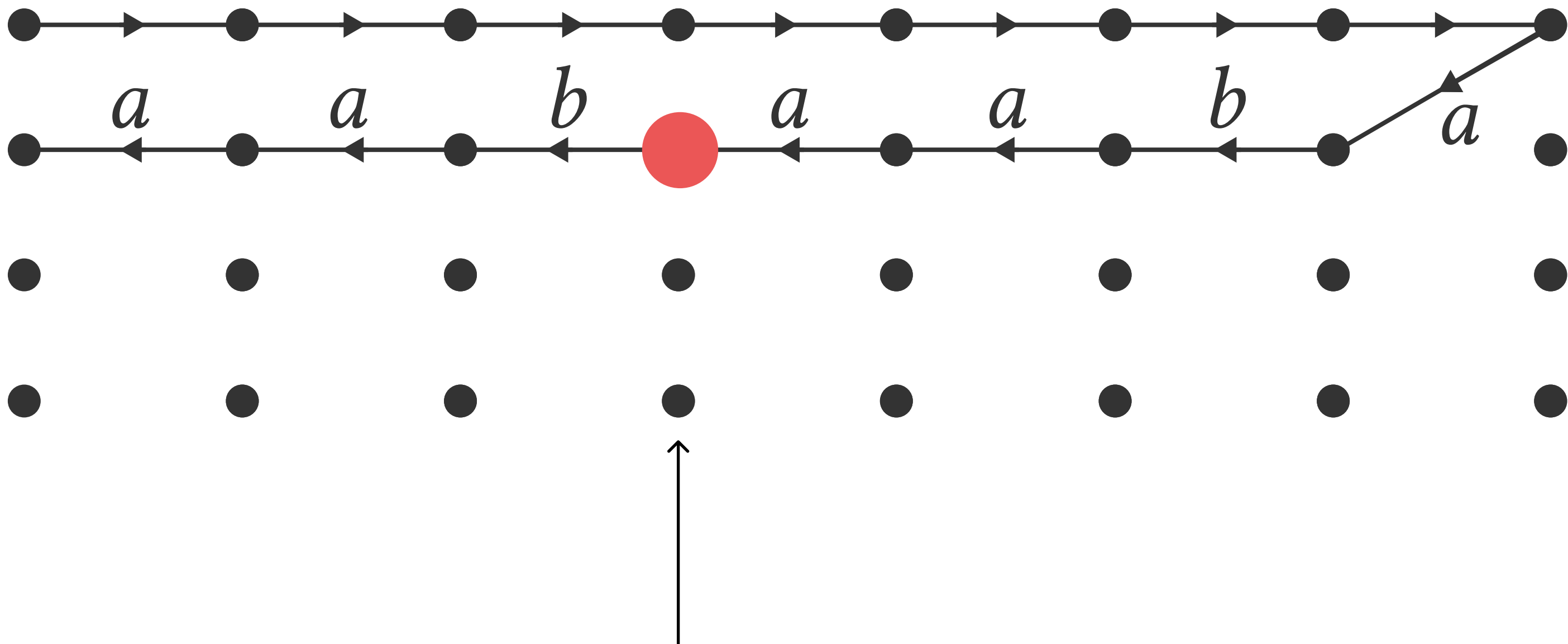
reachable part of
the configuration
graph



string
representation

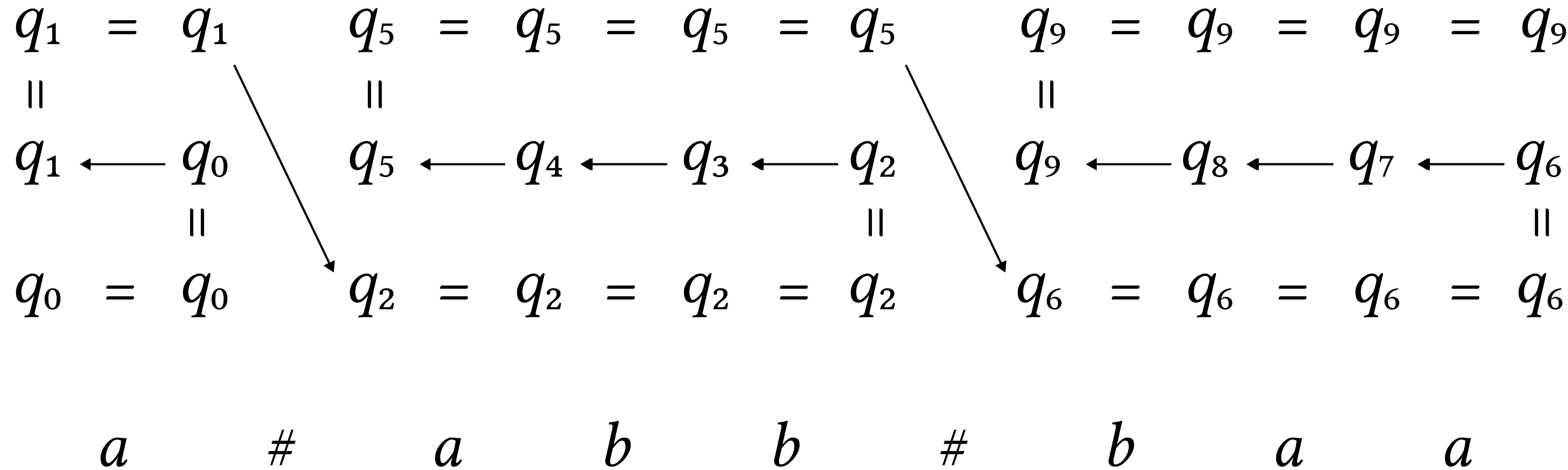


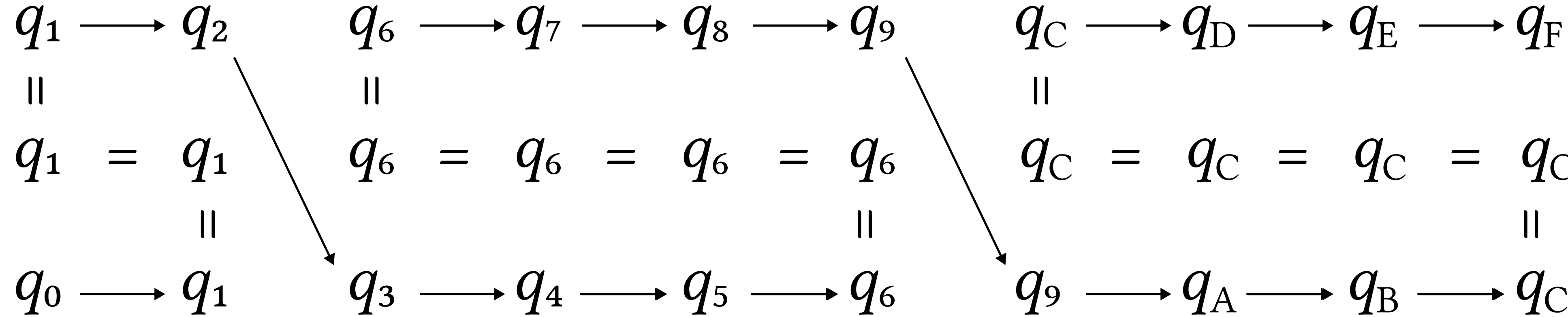


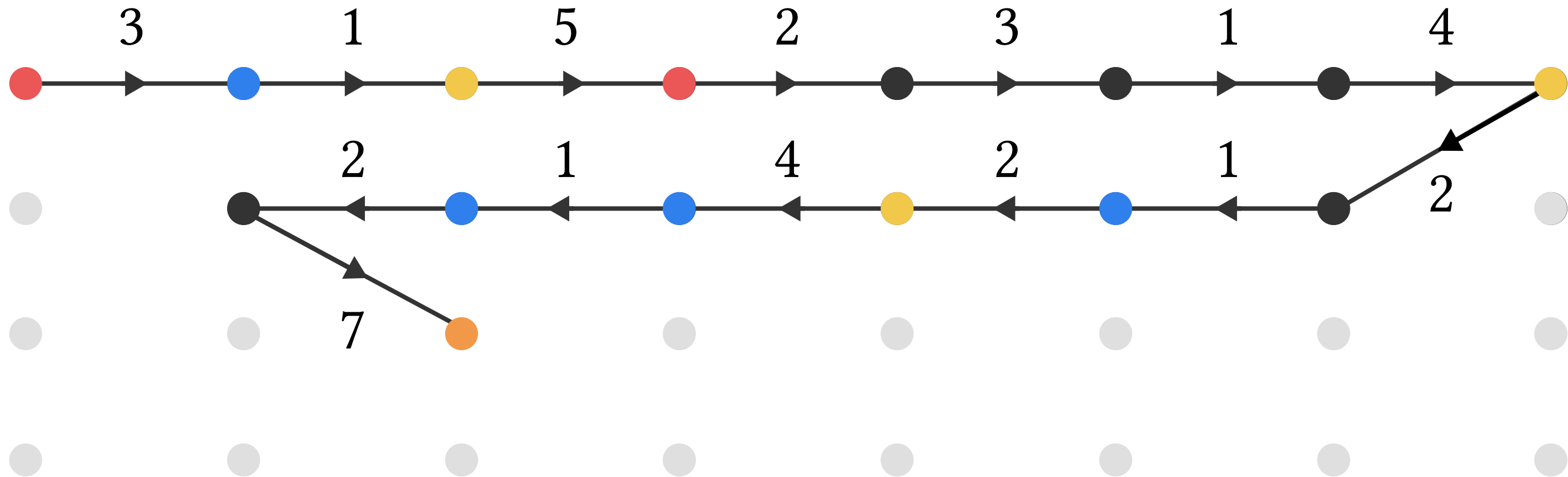


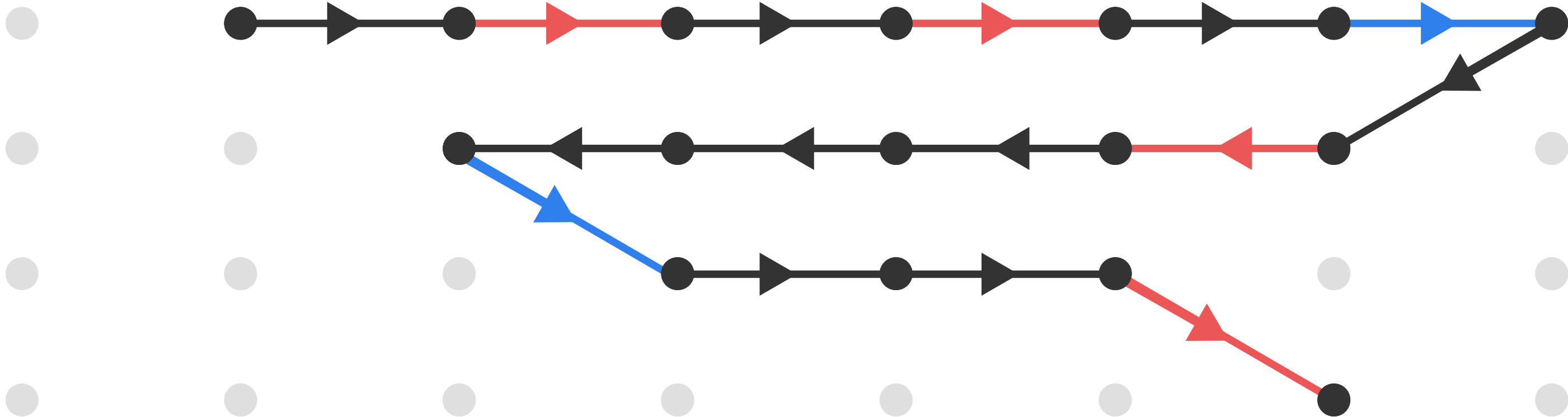
product automaton remembers:

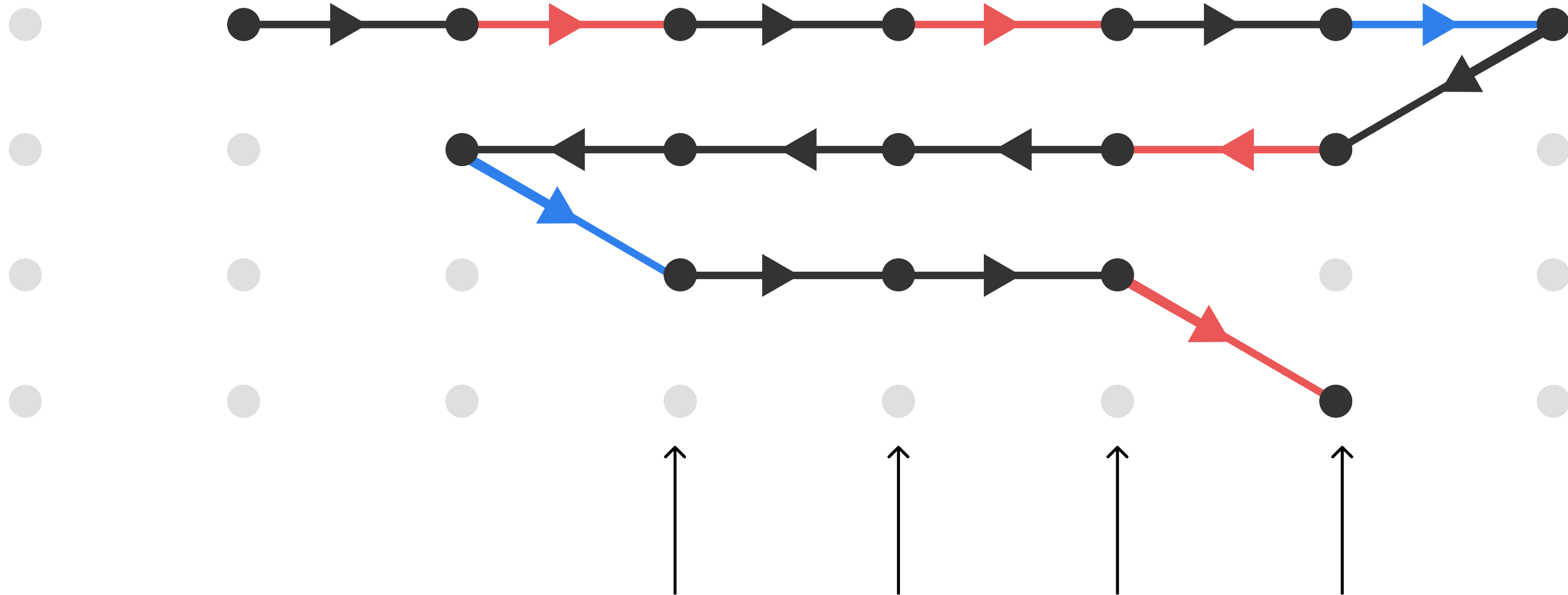
- state of f (red circle); and
- state of g



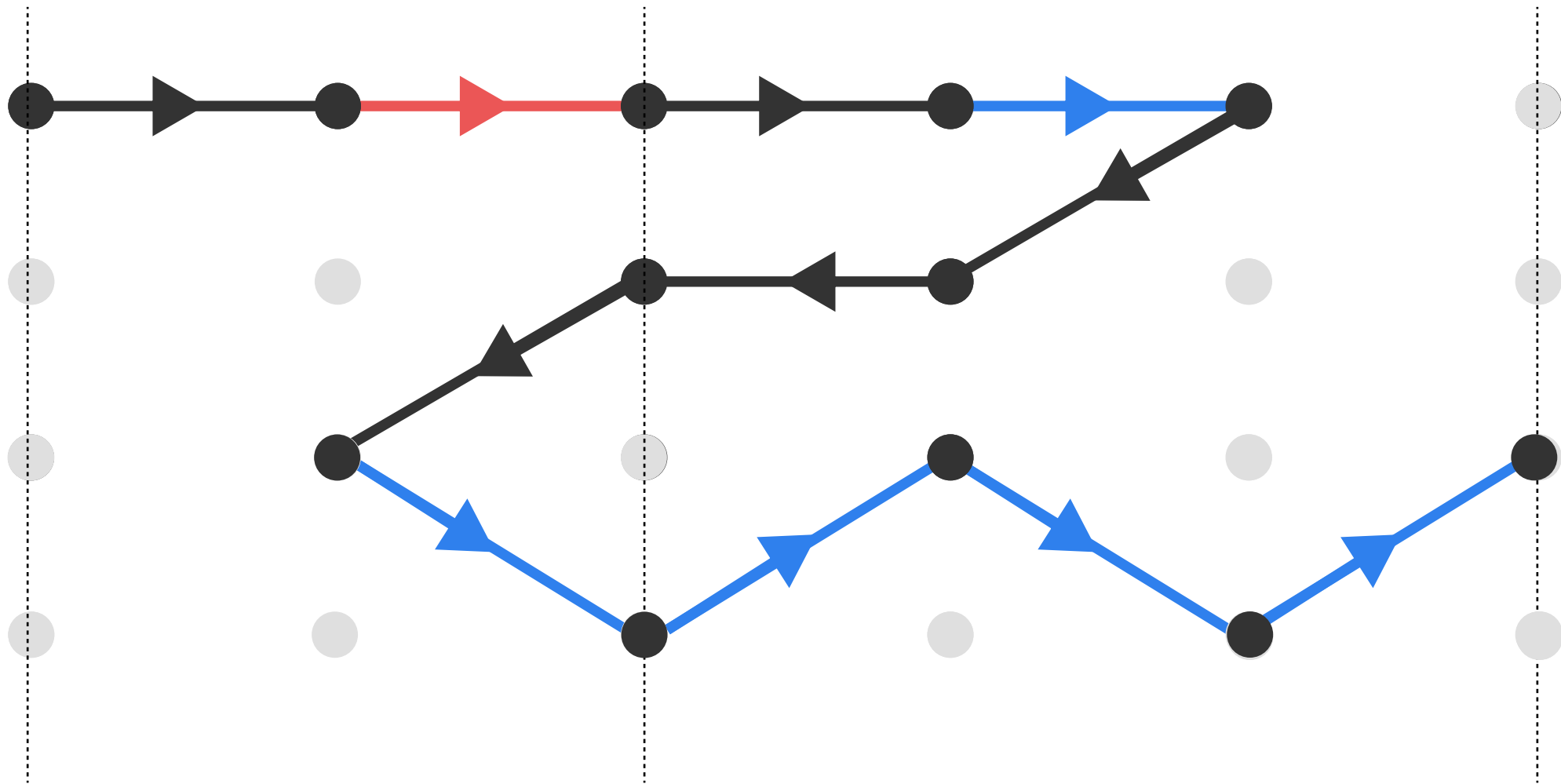


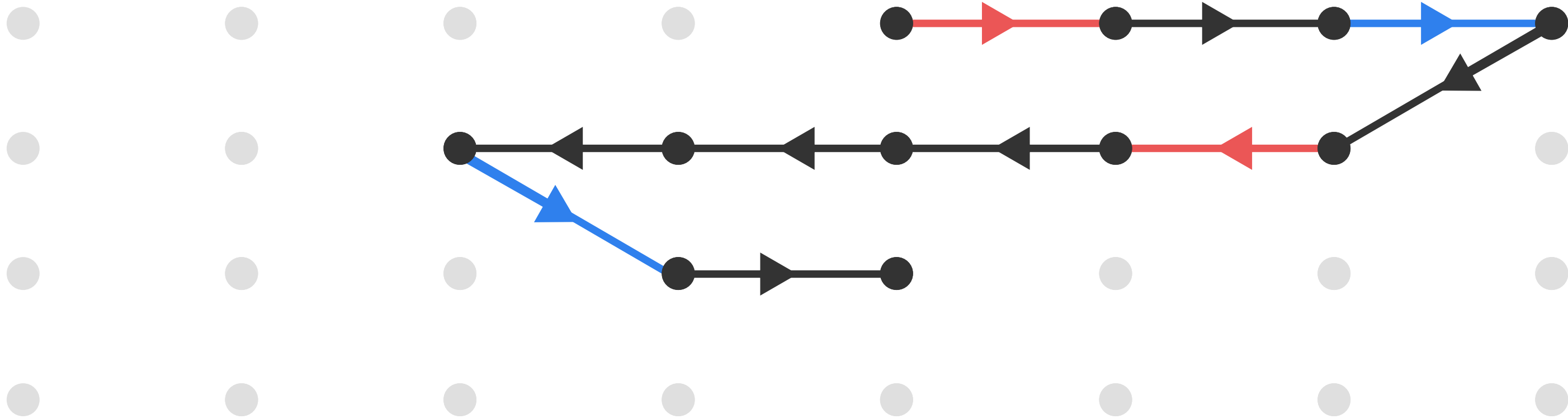






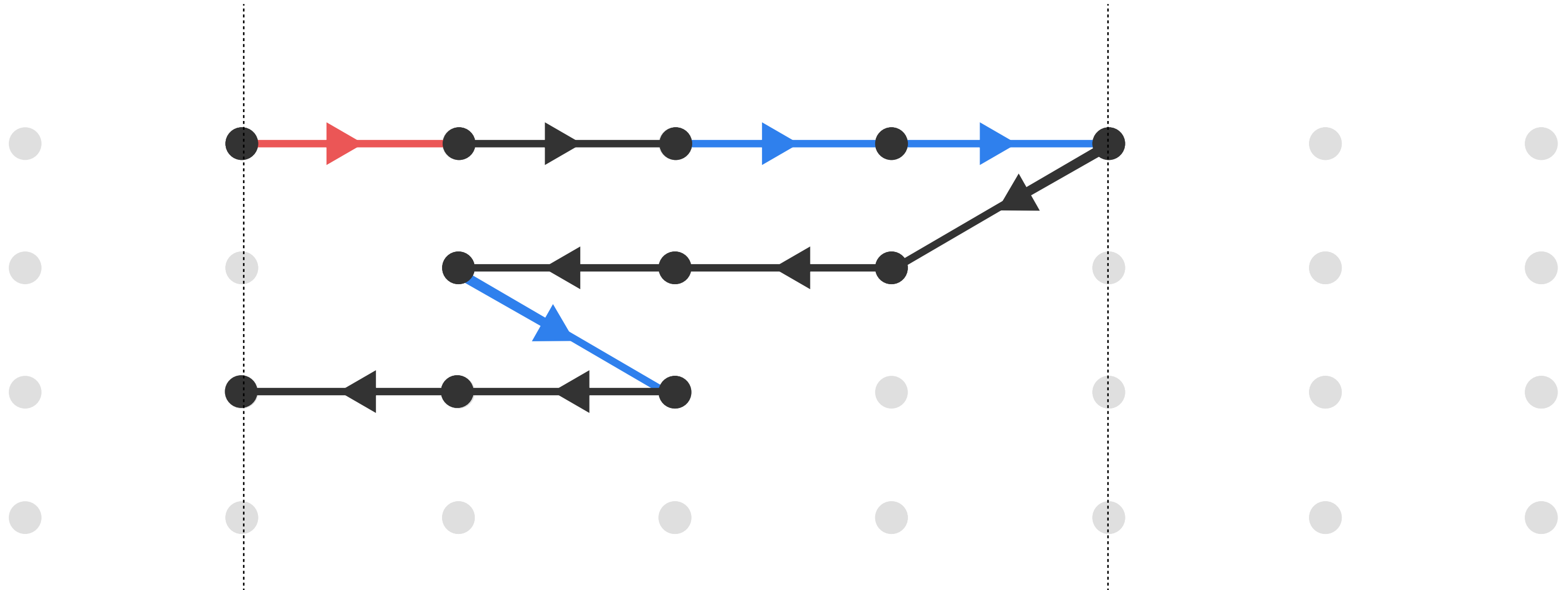
these columns are visited 3 times

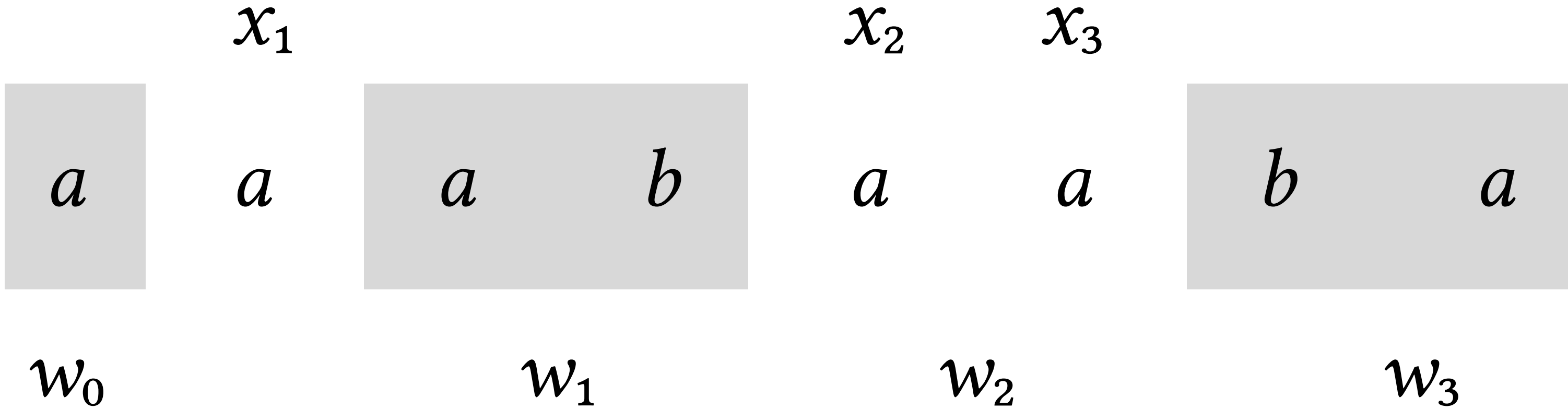
x_{i-1} x_i x_{i+1} 

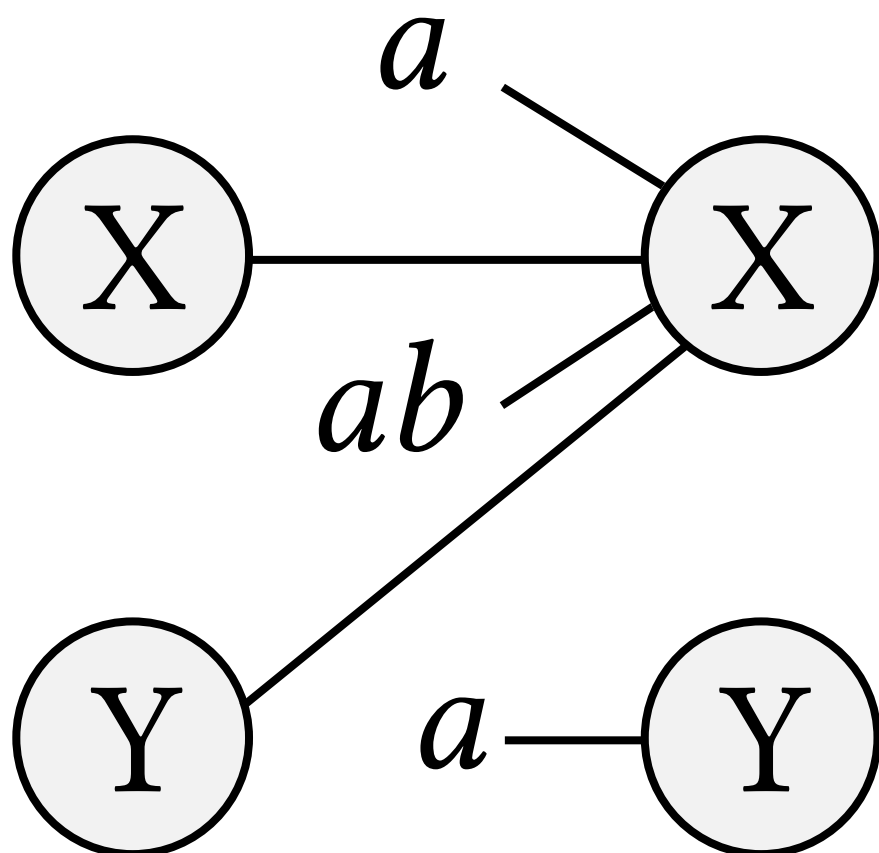


source/target
column

furthest
column

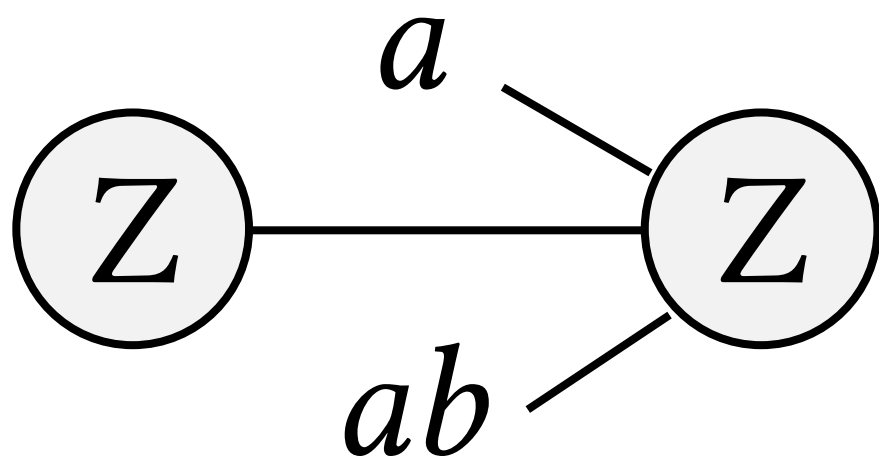




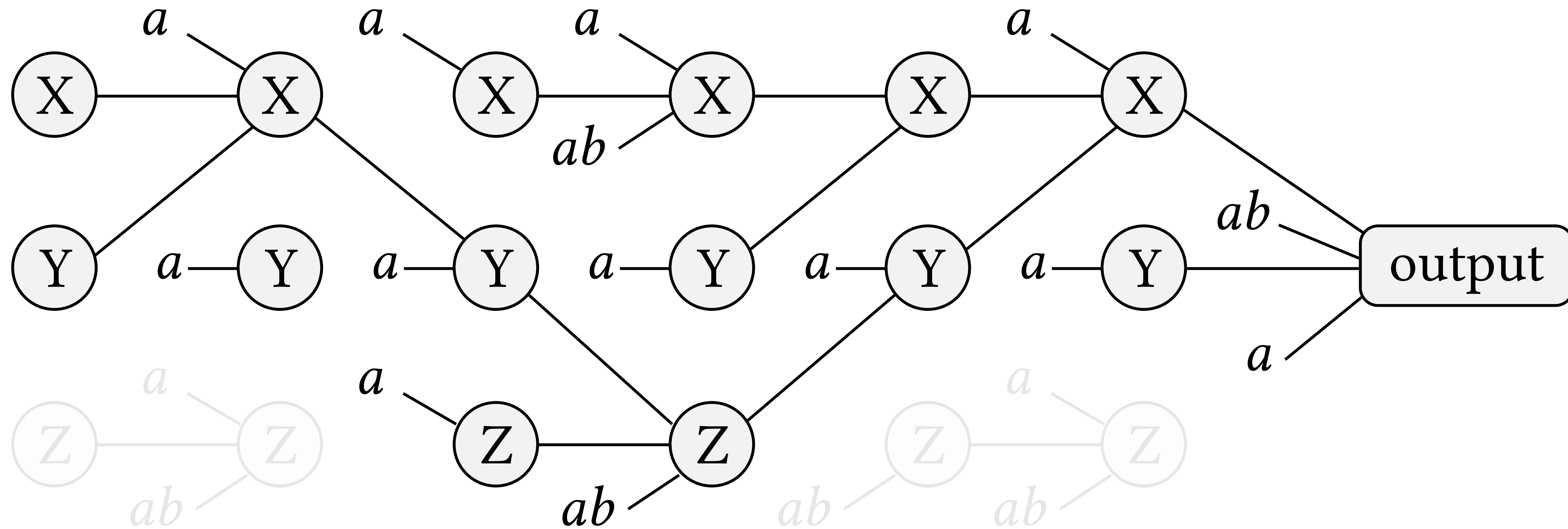


$$X := aXabY$$

$$Y := a$$

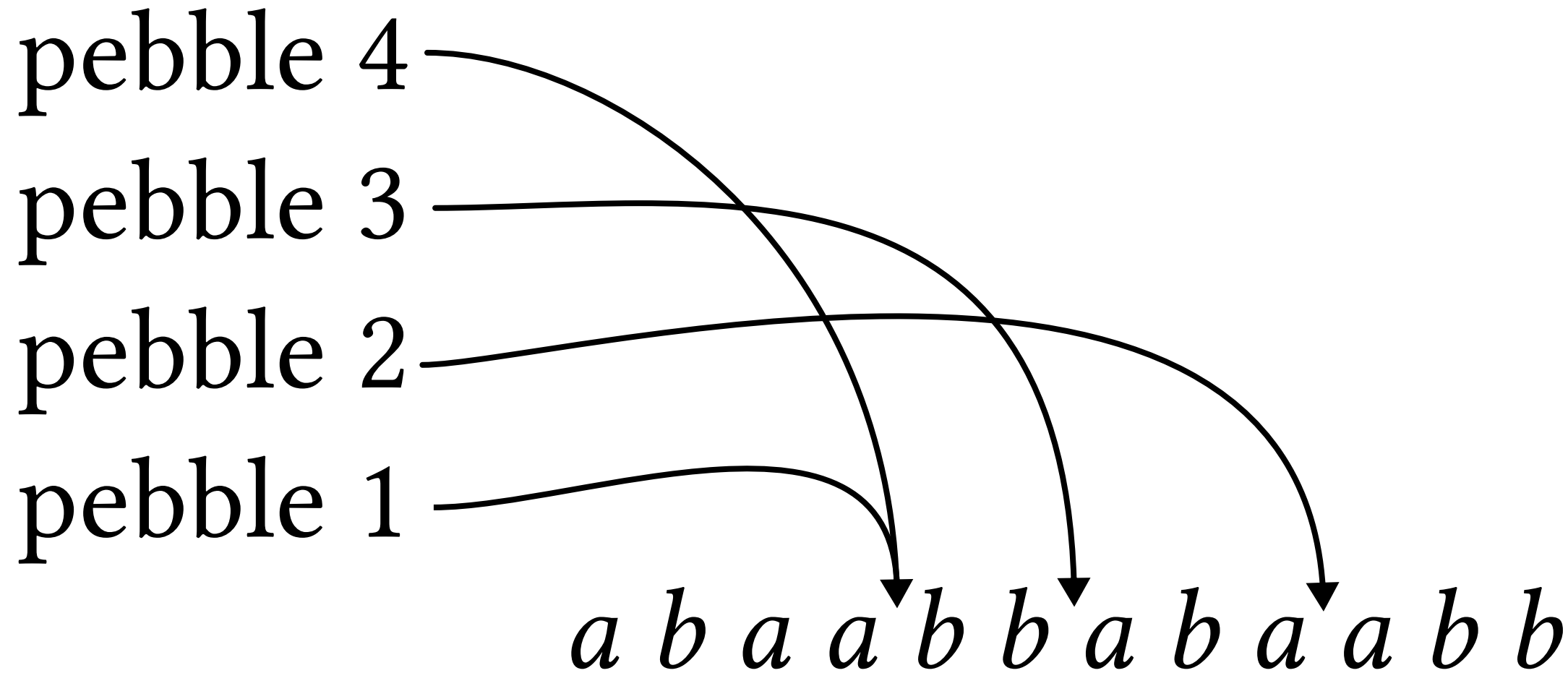


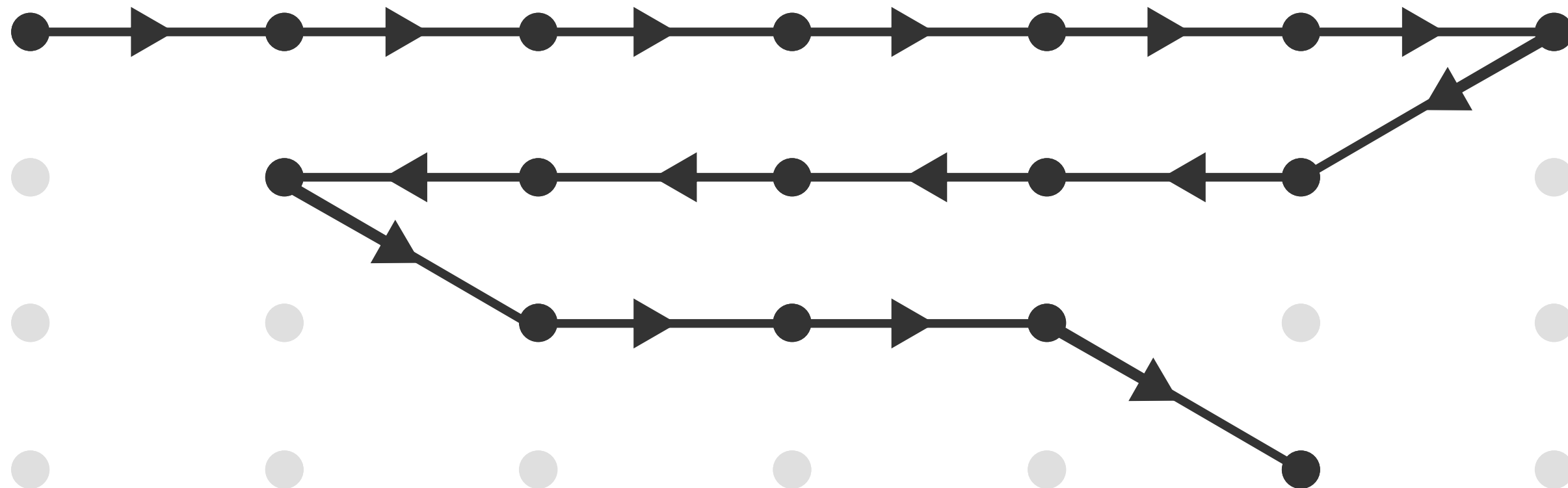
$$Z := aZab$$



$$abcd\#abcd\#abcd\#abcd\#abcd\#abcd\#abcd\# \cdots \#abcd\#$$

The figure displays a sequence of 13 diagrams, each representing a time step t from 0 to 12. Each diagram consists of two vertical arrows. The left arrow is labeled x_1 and the right arrow is labeled x_2 . The relative lengths of these arrows change over time, illustrating the evolution of the system. For example, at $t=0$, the x_1 arrow is short and the x_2 arrow is long. As t increases, the x_1 arrow grows in length while the x_2 arrow shrinks, eventually becoming the longer arrow by $t=12$.





movement of
pebble $l+1$

input string
pebbles $1, \dots, l$

a_1

a_2

a_3

a_4

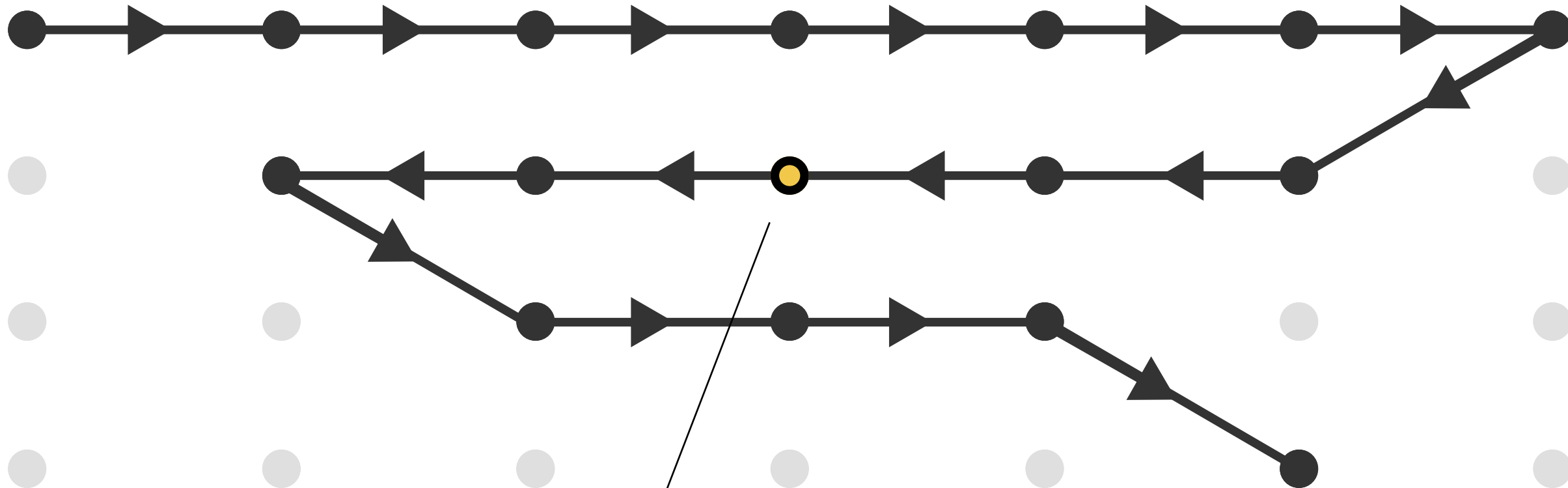
a_5

a_6

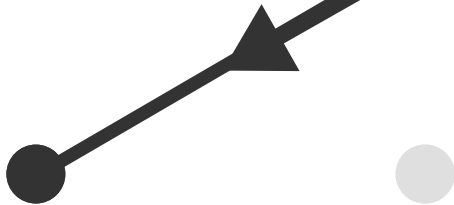
x_1

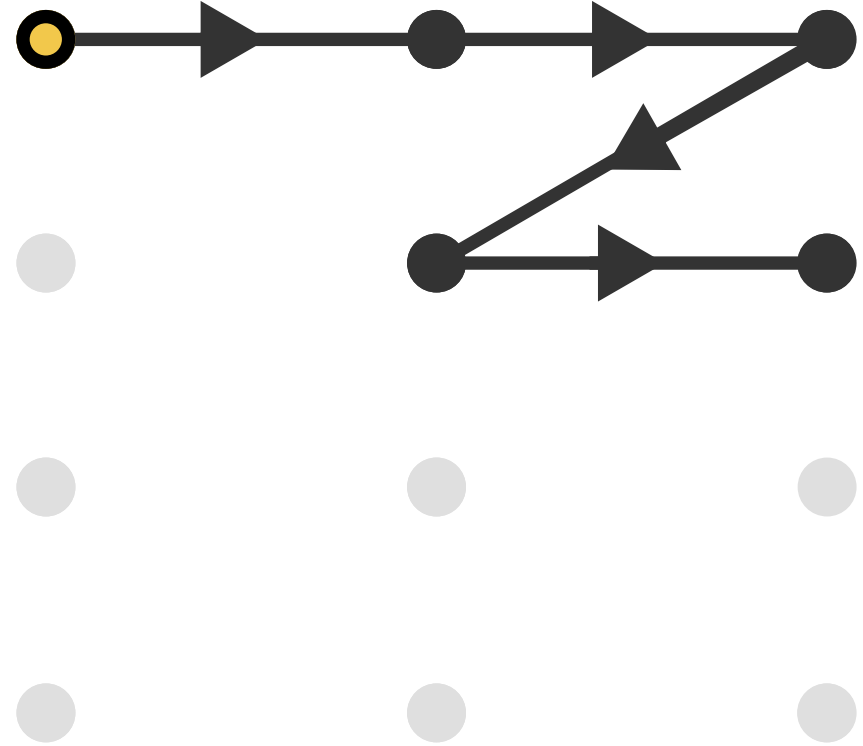
x_2

x_2

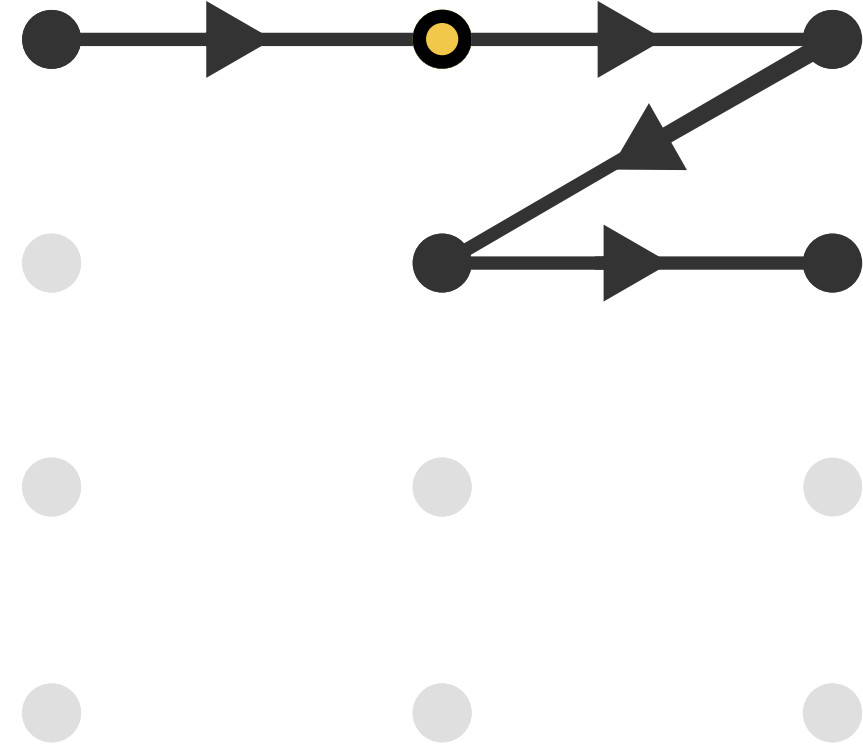


highlighted





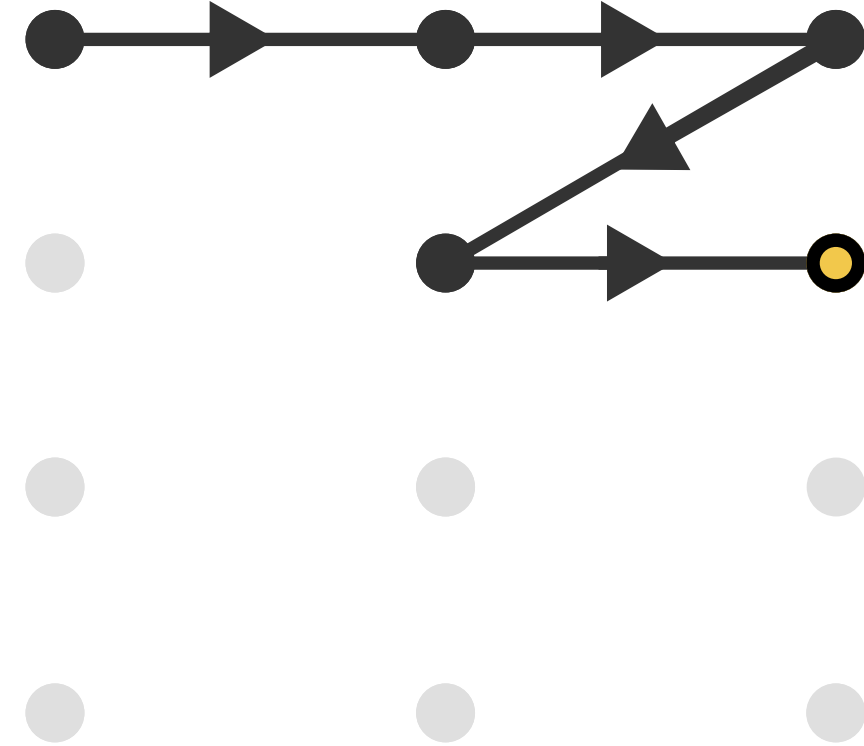
#



#

...

#



balanced run

